MAI 650 — NATURE-INSPIRED COMPUTING

Nature-Inspired Feature Selection for Intrusion Detection Systems

A Comparative Study of Genetic Algorithm, Particle Swarm Optimization, and Ant Colony Optimization

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MOTIVATION

High-Dimensional Traffic Data Slows Down IDS

41

features per connection
in NSL-KDD

78+

flow-level features
in CICIDS2017

2^n-1

candidate subsets to check
for n features — intractable

Feature Selection (FS) keeps only the features that matter — lowering compute cost while preserving (or improving) detection quality.

Nature-inspired metaheuristics (GA, PSO, ACO) search this huge space efficiently, balancing exploration and exploitation — far faster than exhaustive or greedy search.

RESEARCH OBJECTIVES

Five Measurable Objectives (RO1–RO5)

1. Apply GA, PSO, and ACO as wrapper-based feature selectors using the same classifier and pipeline for a fair comparison.
2. Estimate detection rate and false positive rate for each algorithm on NSL-KDD and CICIDS2017.
3. Estimate feature-subset size reduction and its relationship to detection efficiency.
4. Estimate computational cost — convergence time and number of fitness evaluations.
5. Determine which algorithm gives the best accuracy / false-alarm / cost trade-off.

Three Gaps This Project Closes

Gap 1

No Identical-Conditions Comparison

GA has been compared to PSO, and GA to ACO — but no study evaluates all three under one wrapper, classifier, and fitness function.

Gap 2

No Dual-Dataset Evaluation

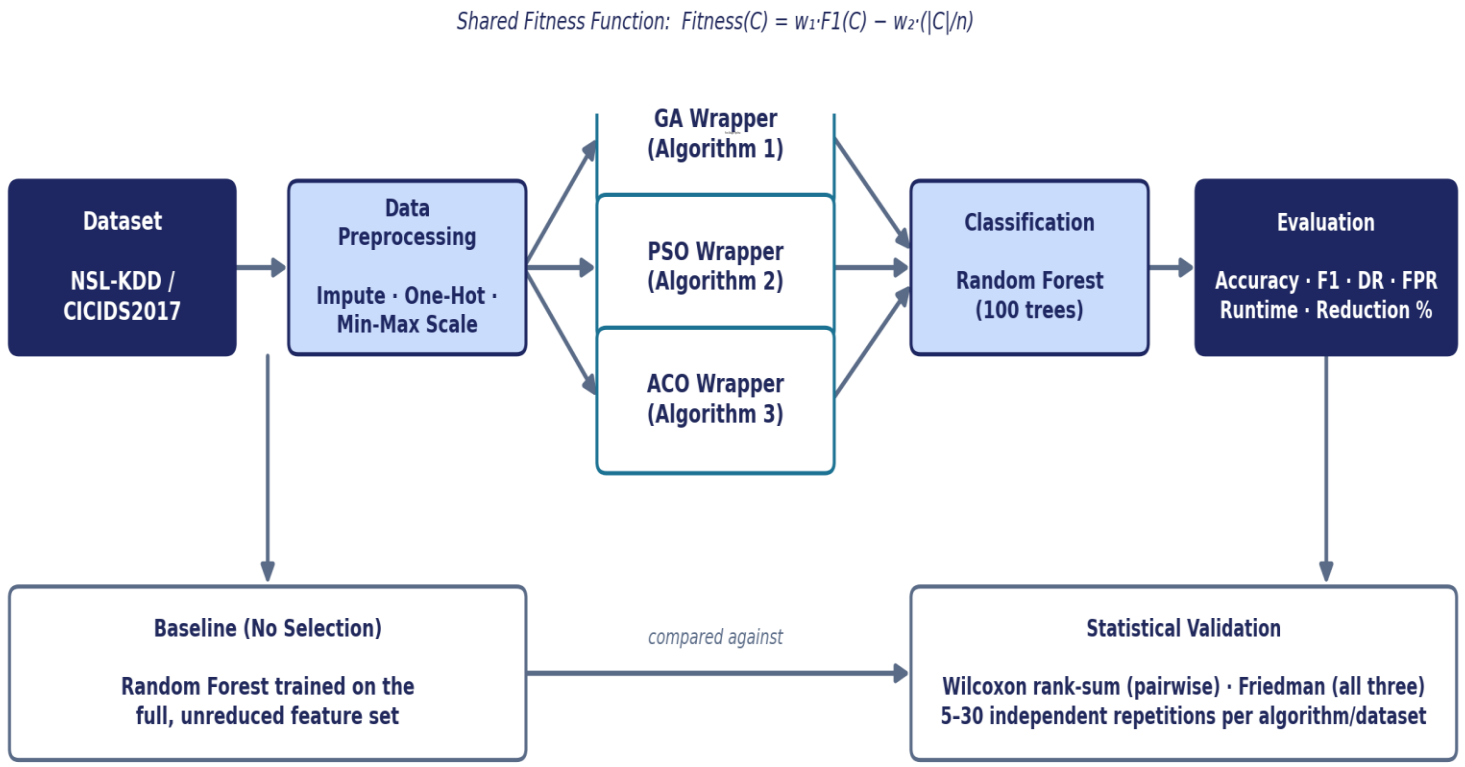
Every reviewed study uses a single benchmark. None evaluates GA, PSO, and ACO on both an established (NSL-KDD) and a modern (CICIDS2017) benchmark.

Gap 3

Fragmented, Single-Metric Reporting

Most papers report accuracy alone. No paper jointly assesses subset size, detection rate, false-positive rate, and computational cost.

A Five-Stage, Reproducible Pipeline

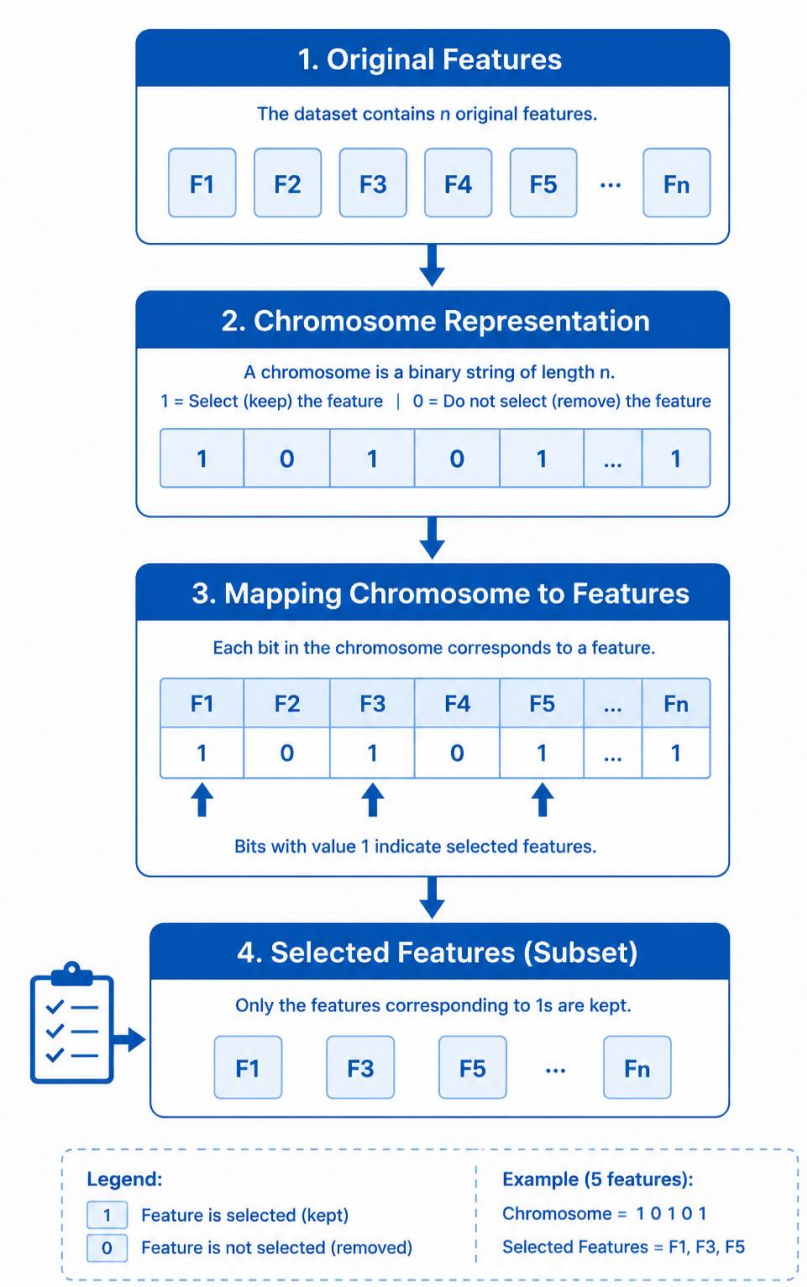


Every algorithm shares the same preprocessing, classifier, and fitness function — isolating the search strategy as the only variable.

Chromosome Representation for Feature Selection

Each bit in the chromosome represents one feature.

- **1 = Feature is selected (kept).**
- **0 = Feature is not selected (discarded).**
- **Only the selected features are used to train the Random Forest classifier.**



Three Different Search Strategies, One Fitness Function

GA

Discrete Recombination

Representation

Binary chromosome
(1 bit = 1 feature)

Core operators

Selection → Crossover → Mutation

Natural fit for the binary include/exclude decision.

PSO

Continuous Velocity Search

Representation

Real-valued position,
sigmoid-binarized

Core operators

pbest / gbest velocity update

Fewer control parameters; needs discretization.

ACO

Pheromone-Guided Traversal

Representation

Feature graph,
ant path = subset

Core operators

Pheromone deposit + evaporation

Collective memory across iterations.

Shared fitness function: $Fitness(C) = w_1 \cdot F1(C) - w_2 \cdot (|C|/n)$ $(w_1 = 0.9, w_2 = 0.1)$

EXPERIMENTAL SETUP

Datasets, Parameters, and Metrics

NSL-KDD

41 features · 125,973 train / 22,544 test
DoS, Probe, R2L, U2R + normal

CICIDS2017

78+ flow features · CICFlowMeter
DDoS, DoS, Brute Force, Web Attack, Infiltration, Botnet, Port Scan

Parameter	GA	PSO	ACO
Population / swarm / colony	50	50	50
Generations / iterations	100	100	100
Key parameters	pc=0.8, pm=0.02	w: 0.9→0.4, c1=c2=2.0	ρ=0.1, α=1.0, β=2.0

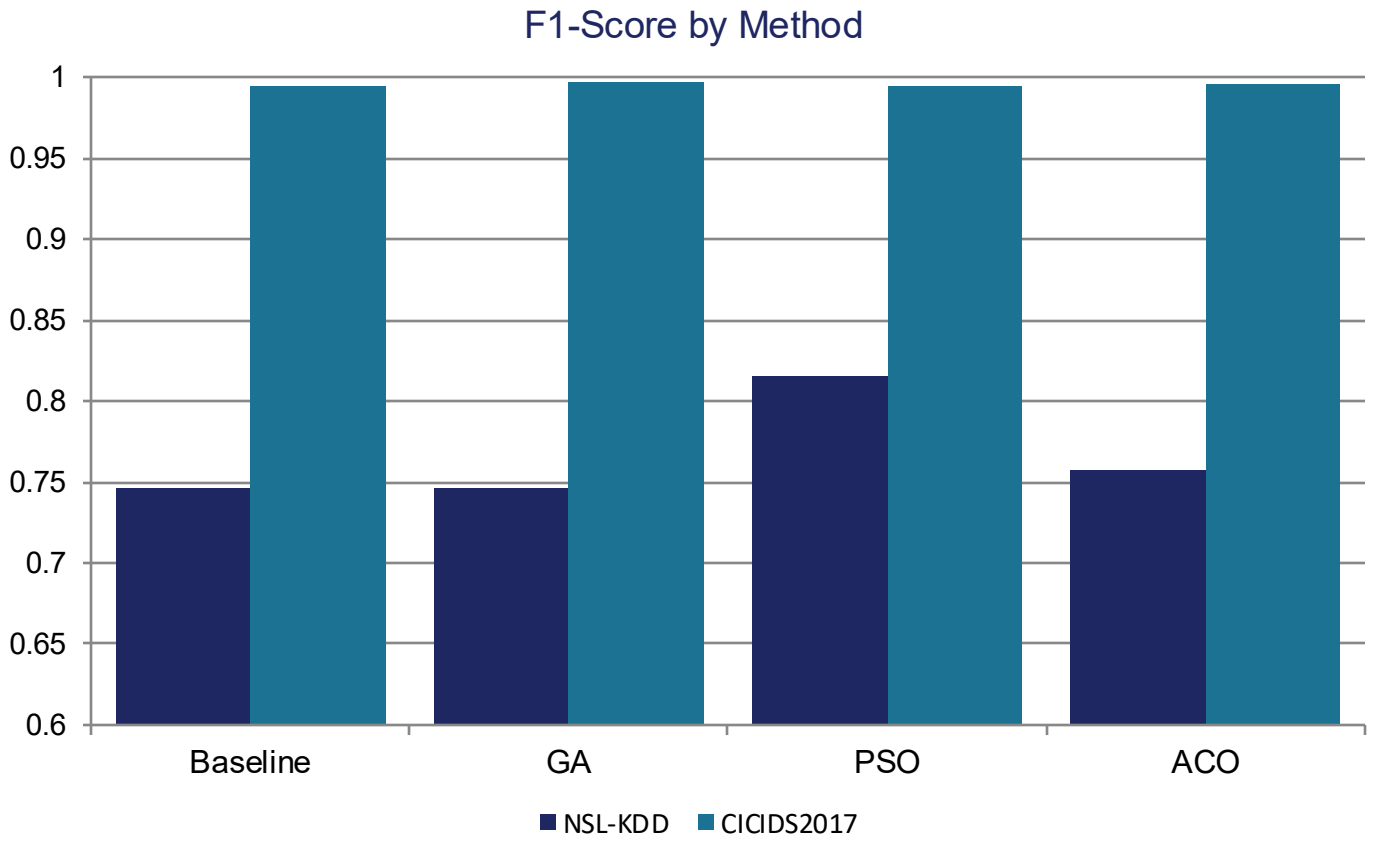
Random Forest classifier (100 trees, Gini) shared across all algorithms and the baseline.

Metrics: Accuracy · Precision · Recall (Detection Rate) · F1 · False Positive Rate · Feature Reduction % · Runtime

Statistical validation: Wilcoxon rank-sum (pairwise) and Friedman (all three) tests on F1-score across independent runs.

RESULTS — PILOT CONFIGURATION

Feature Selection Preserves — or Improves — Detection Quality



43–80%

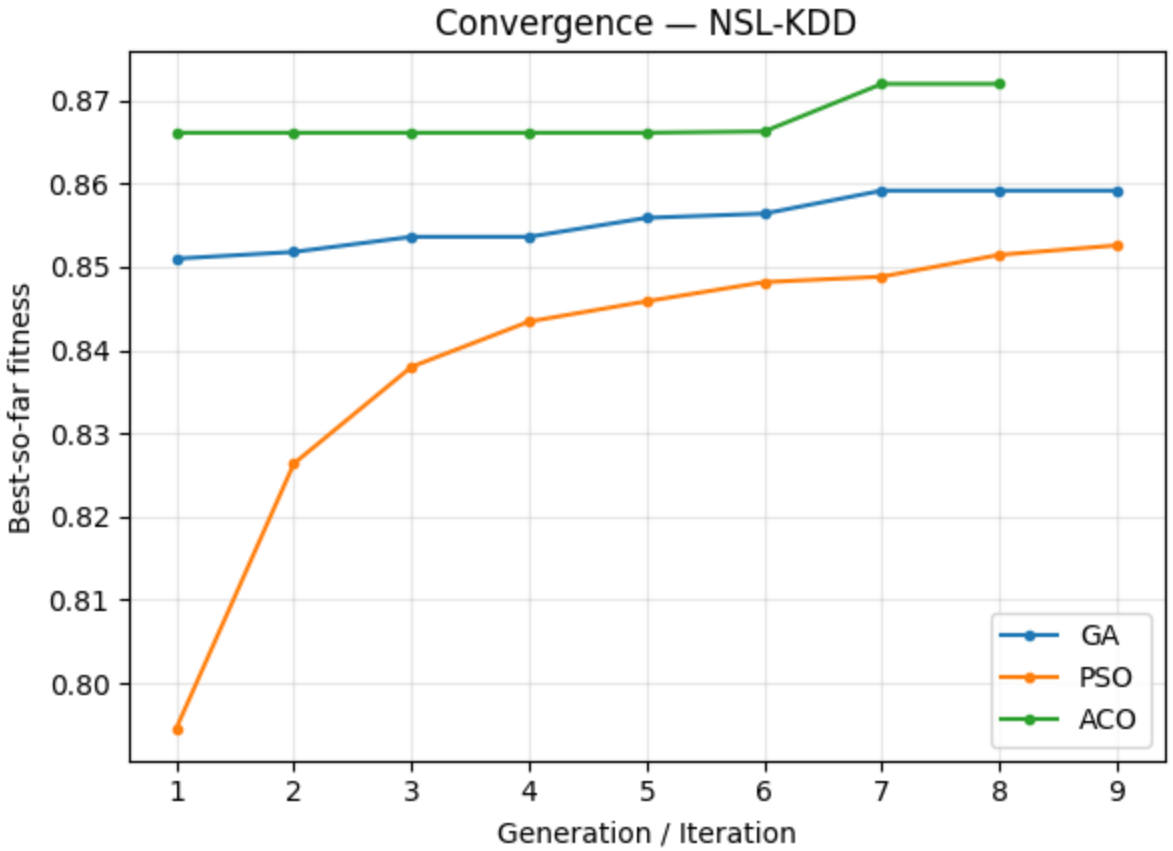
of original features removed across GA / PSO / ACO, on both datasets

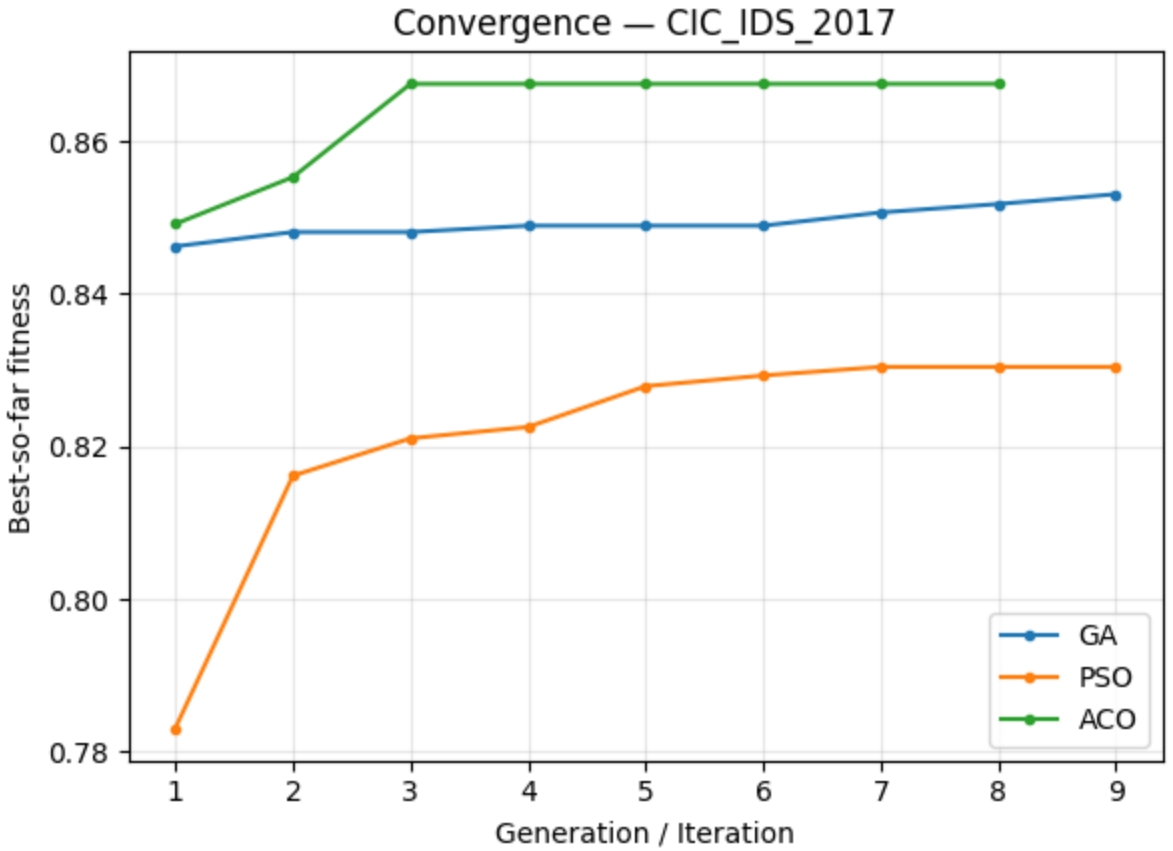
GA on CICIDS2017

F1 0.9974 vs. 0.9952 baseline

— beats the full 78-feature baseline while using 63% fewer features.

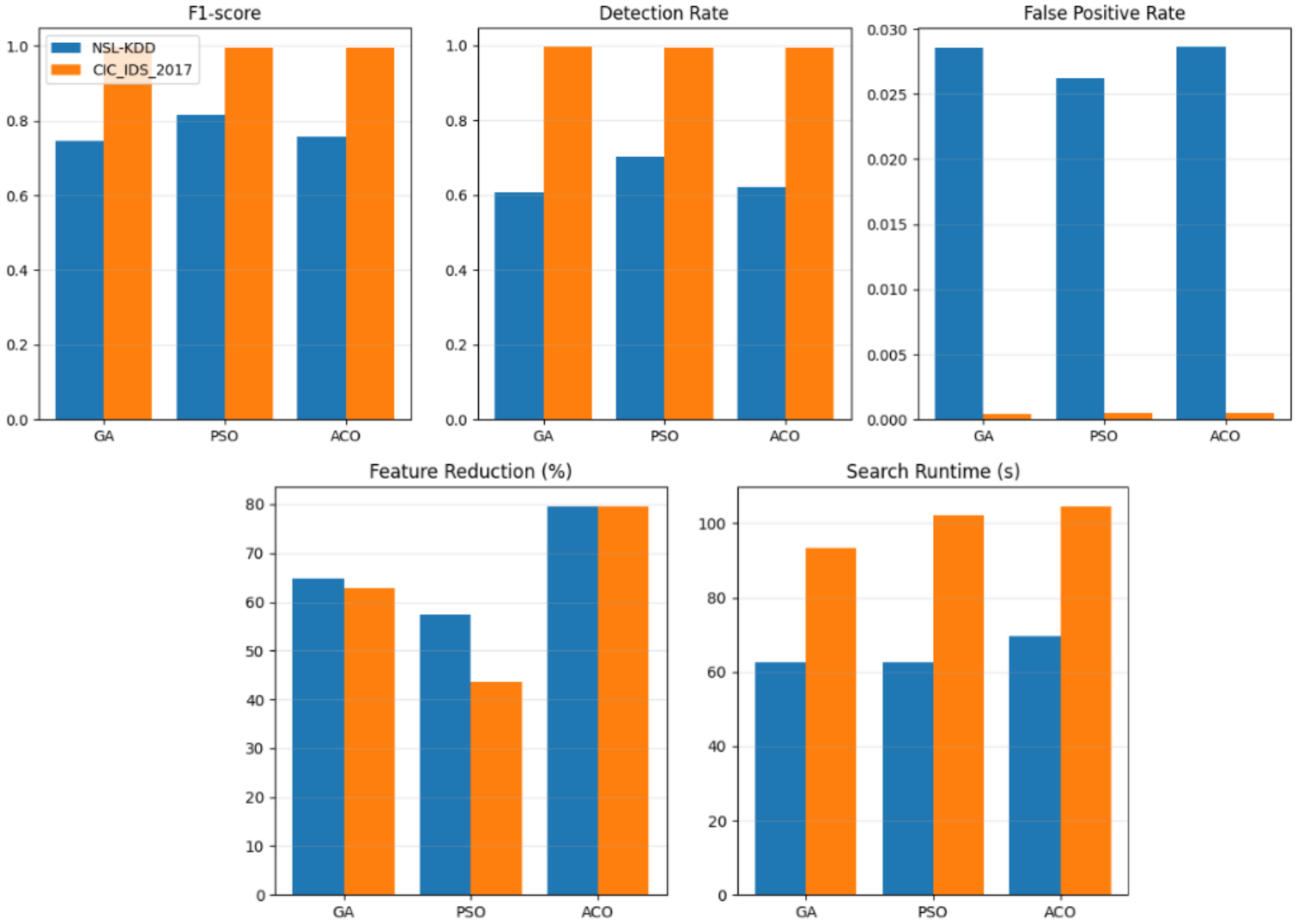
ACO Converges Fastest — But Not Fastest in Wall-Clock Time





ACO plateaus within 2–3 iterations (pheromone reinforcement), but its per-iteration cost is higher by a factor of d — so **GA/PSO often finish sooner in wall-clock time.**

ACO Shrinks Features Most; PSO Wins on the Harder Dataset



PSO leads F1 & Detection Rate on NSL-KDD (0.816 / 0.703) · ACO removes the most features on both datasets (~80%) · CICIDS2017 stays >99% across the board

STATISTICAL VALIDATION (RO5)

No Significant Difference — Yet, at Pilot Sample Size

Dataset	Comparison	p-value	Result (α=0.05)
NSL-KDD	GA vs. PSO	0.6250	not significant
NSL-KDD	PSO vs. ACO	0.1250	not significant
NSL-KDD	Friedman (all 3)	0.2466	not significant
CICIDS2017	Friedman (all 3)	0.5488	not significant

Why this matters

- Only 5 repetitions in this pilot — Wilcoxon has limited power at this sample size.
- Section IV-F specifies 30 repetitions for the full-budget run.
- Reported transparently rather than omitted — consistent with jointly reporting every metric, not just favorable ones.

DISCUSSION

Findings Mapped Back to the Research Objectives

RO1	Fair comparison achieved by construction — identical classifier, pipeline, fitness function.
RO2 / RO3	Table VII reports DR, FPR, and subset size jointly — closing Gap 3.
RO4	Convergence (iterations) and wall-clock runtime don't always agree — ACO fastest to converge, not fastest overall.
RO5	PSO best for accuracy-critical NSL-KDD-like settings; ACO best when minimizing deployed features matters most.

A Reproducible Foundation for the Full-Budget Run

Closed 3 research gaps:

identical-conditions GA/PSO/ACO comparison, on both NSL-KDD and CICIDS2017, with jointly-reported metrics.

Pilot results:

43–80% feature reduction with preserved or improved detection quality; ACO smallest subsets, PSO strongest on NSL-KDD.

Next steps:

full-budget run (population 50 / 100 generations, 30 statistical repetitions), S-shaped vs. V-shaped PSO transfer functions, ACO ρ/α/β sensitivity analysis.

Thank You

Questions?

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